Riphah HireHive

1. System Features

1. User Registration and Login
2. User Management and Roles
3. Profile and CV Management
4. Event Management
5. FYP Team and Project Registration
6. Volunteer Registration and Selection
7. Company Registration and Invitation
8. Table Allocation and Tags
9. Seminar and Attendance
10. LCD Issue and Return
11. QR Code and Check-in
12. Job Fair and Interview Management
13. FYP Exhibition and Evaluation
14. Volunteer Task Management and Tracking
15. Search and AI Matching
16. Notifications and Announcements
17. Certificates
18. Analytics and Reports
19. Alumni Registration and Login
20. Student Selection for Mock Interview
21. Mock Interview Notifications
22. Alumni Dashboard and Student CV Access
23. Alumni Feedback and Review
24. WhatsApp Message Intake
25. Requirement-Based Student Suggestion

2. Software Requirements

1. User Registration and Login

- SRS-1.1: The system shall allow FYP students, volunteers, and 2nd to 6th semester students to register using a valid Riphah email address (for example 54910@students.riphah.edu.pk).
- SRS-1.2: The system shall reject a student or volunteer registration when the email address is not in the Riphah format. Alumni are exempt from this check.
- SRS-1.3: The system shall verify that the email address is unique, and for students and volunteers that the SAP ID is also unique, before creating an account.
- SRS-1.4: The system shall send a verification code to the registered email address.
- SRS-1.5: The system shall not allow a student or volunteer to log in before completing account verification.
- SRS-1.6: The system shall encrypt user passwords before storing them in the database.
- SRS-1.7: The system shall allow alumni to register through a separate alumni registration link using only an email address and a password (see feature 19).

2. User Management and Roles

- SRS-2.1: The system shall allow the admin team to create, edit, deactivate, and delete user accounts.
- SRS-2.2: The system shall show each user only the screens and data that are allowed for their role.
- SRS-2.3: The system shall allow the event organizer to give selected team members the right to edit shared forms such as the demand form.
- SRS-2.4: The system shall generate a username and password for each selected volunteer and send it to them.
- SRS-2.5: The system shall provide an alumni role that can open only the alumni dashboard, the details of the students selected for the mock interview, and the feedback screens.
- SRS-2.6: The system shall allow only the admin team to open the WhatsApp requirement records, the sender details, and the student suggestions (see features 24 and 25).

3. Profile and CV Management

- SRS-3.1: The system shall allow users to update their name, phone number, and profile picture.
- SRS-3.2: The system shall allow users to change their password after verifying their existing password.
- SRS-3.3: The system shall allow FYP students and 2nd to 6th semester students to upload and replace their CV.
- SRS-3.4: The system shall display an FYP student's name, email, SAP ID, project details, and supervisor on their portal.
- SRS-3.5: The system shall tell a student when required profile information or the CV is missing.

4. Event Management

- SRS-4.1: The system shall allow the admin team to create an open house event with its date, time, venue, and registration deadlines.

- SRS-4.2: The system shall provide registration links for FYP teams, volunteers, students, and companies.
- SRS-4.3: The system shall provide a demand form for tables, chairs, sound system, microphones, and decoration theme.
- SRS-4.4: The system shall export the approved demand form as a PDF.
- SRS-4.5: The system shall collect each team member's free and busy time and suggest a common free slot for meetings.
- SRS-4.6: The system shall treat the mock interview as part of the same open house project, arranged one week before the open house, and link it to the open house event.

5. FYP Team and Project Registration

- SRS-5.1: The system shall allow FYP students to submit team details, including the number of members, member names, and SAP IDs.
- SRS-5.2: The system shall allow FYP students to submit the supervisor name, technology used, and project description.
- SRS-5.3: The system shall prevent the same SAP ID from being registered in more than one team.
- SRS-5.4: The system shall warn the team when required information, such as the supervisor name, is missing.
- SRS-5.5: The system shall allow supervisors to approve FYP team registrations.
- SRS-5.6: The system shall allow the admin team to view and export the list of registered teams by program (BCA, SE, CS).

6. Volunteer Registration and Selection

- SRS-6.1: The system shall allow students to register as volunteers by entering their name, SAP ID, WhatsApp number, and experience.
- SRS-6.2: The system shall not accept volunteer registrations after the deadline.
- SRS-6.3: The system shall shortlist volunteers based on experience and participation in previous open houses.
- SRS-6.4: The system shall suggest suitable volunteers for each duty, such as social media or welcoming.
- SRS-6.5: The system shall display each selected volunteer's duties, placement, responsibilities, and rules on their portal.

7. Company Registration and Invitation

- SRS-7.1: The system shall allow the admin team to add and store company records, including company name, contact person, email, and phone number.
- SRS-7.2: The system shall send an invitation email with the invitation card, student information, table details, and attachments.
- SRS-7.3: The system shall allow companies to confirm attendance with the number of people and the name of the team leader.

8. Table Allocation and Tags

- SRS-8.1: The system shall assign a unique table number to every project.

- SRS-8.2: The system shall assign table numbers in one continuous sequence across programs (for example, if BCA uses tables 1 to 8, SE starts at table 9).
- SRS-8.3: The system shall prevent two projects from having the same table number.
- SRS-8.4: The system shall generate a printable table tag containing the project name, SAP ID, email, supervisor, technology, program, and table number.
- SRS-8.5: The system shall allow the admin team to review and correct a tag before printing.
- SRS-8.6: The system shall display the table number and location to each FYP team on their portal.

9. Seminar and Attendance

- SRS-9.1: The system shall allow the admin team to schedule a seminar with a room number and time.
- SRS-9.2: The system shall notify students, volunteers, teachers, and the admin team of the seminar room and time on their portals.
- SRS-9.3: The system shall generate an attendance sheet for registered FYP students and volunteers with SAP ID, name, and role.
- SRS-9.4: The system shall allow the admin team to mark seminar attendance.
- SRS-9.5: The system shall notify students who did not attend the seminar that they are not eligible for the open house.
- SRS-9.6: The system shall allow the admin team to export the attendance sheet.

10. LCD Issue and Return

- SRS-10.1: The system shall allow FYP groups to request an LCD from the lab.
- SRS-10.2: The system shall fill the group name, member names, SAP IDs, and supervisor name in the LCD request from the project record.
- SRS-10.3: The system shall display the disclaimer that the group bears any damage, and shall require the group to accept it before the request is submitted.
- SRS-10.4: The system shall show lab staff a list of all LCD requests.
- SRS-10.5: The system shall allow lab staff to record the issue and return of the LCD, HDMI cable, and power cable, with a note for any damage.
- SRS-10.6: The system shall keep a record of LCD issue and return for every open house and list the groups that have not returned the LCD.

11. QR Code and Check-in

- SRS-11.1: The system shall generate a unique QR code for each invited company.
- SRS-11.2: The system shall allow volunteers at the welcome desk to check in a company by scanning its QR code or ticking its checkbox.
- SRS-11.3: The system shall allow volunteers to search for a company by company name or person name.
- SRS-11.4: The system shall show volunteers the list of expected companies with company name, number of people, and main person.
- SRS-11.5: The system shall allow volunteers to add a new entry for a company that is not registered and mark it present.
- SRS-11.6: The system shall save every check-in record and allow it to be exported.

12. Job Fair and Interview Management

- SRS-12.1: The system shall allow companies to search CVs by keyword, skill, or program.
- SRS-12.2: The system shall show whether a student's skills and experience match what the company is looking for.
- SRS-12.3: The system shall allow companies to shortlist students.
- SRS-12.4: The system shall allow companies to schedule interviews with shortlisted students.
- SRS-12.5: The system shall notify students about their interview schedule.

13. FYP Exhibition and Evaluation

- SRS-13.1: The system shall allow companies and guests to browse FYP projects and filter them by technology or program.
- SRS-13.2: The system shall allow companies to save a shortlist of projects.
- SRS-13.3: The system shall allow FYP teams to book a presentation room and time slot.

14. Volunteer Task Management and Tracking

- SRS-14.1: The system shall allow the admin team to assign duties and tasks to volunteers with a priority.
- SRS-14.2: The system shall show each volunteer a to-do checklist and allow them to mark tasks as done.
- SRS-14.3: The system shall remind volunteers about tasks that are not yet completed.
- SRS-14.4: The system shall allow volunteers to check in and check out, and record the time.
- SRS-14.5: The system shall allow the admin team to view volunteers' placement and task progress, and to send reminder messages to them.
- SRS-14.6: The system shall allow volunteers to submit a short report of their work and allow the admin team to give appreciation for good performance.

15. Search and AI Matching

- SRS-15.1: The system shall allow companies to enter the kinds of projects they are interested in.
- SRS-15.2: The system shall recommend matching projects and their table numbers to a company based on its interests.
- SRS-15.3: The system shall allow users to search FYP projects by keyword, technology, or program.
- SRS-15.4: The system shall flag duplicate or invalid records, such as the same SAP ID twice, a wrong email domain, or a missing supervisor, for admin review.
- SRS-15.5: The system shall allow companies to mark a project as "interested" or "interview later".

16. Notifications and Announcements

- SRS-16.1: The system shall send notifications to users inside the application and by email.
- SRS-16.2: The system shall send reminders about deadlines, the seminar, and the event day.
- SRS-16.3: The system shall allow the admin team to send one notification to a whole group, such as all volunteers, all FYP students, or all companies.

- SRS-16.4: The system shall allow the admin team to send announcements on the event day, such as a room change.
- SRS-16.5: The system shall allow users to manage their notification preferences.
- SRS-16.6: The system shall keep a history of notifications sent.
- SRS-16.7: The system shall send in-app and email notifications to students selected for the mock interview, as described in feature 21.
- SRS-16.8: The system shall remind selected students who have not yet uploaded their CV.
- SRS-16.9: The system shall notify the admin team on the dashboard when a new WhatsApp requirement record is received or when its sender details are missing.

17. Certificates

- SRS-17.1: The system shall generate an e-certificate or experience letter for volunteers after the event.
- SRS-17.2: The system shall base a volunteer's certificate on their recorded attendance and duty hours.
- SRS-17.3: The system shall allow the admin team to approve certificates before they are released.
- SRS-17.4: The system shall allow users to download their certificates from their portal.
- SRS-17.5: The system shall keep a record of all certificates issued.

18. Analytics and Reports

- SRS-18.1: The system shall display a dashboard of registrations, volunteers selected, tables assigned, companies confirmed, and attendance.
- SRS-18.2: The system shall alert the admin team about problems, such as companies not yet confirmed or tables with no project.
- SRS-18.3: The system shall allow the admin team to export lists and reports to Excel and PDF.
- SRS-18.4: The system shall generate a post-event report of attendance, companies visited, and volunteer performance.
- SRS-18.5: The system shall keep the data of previous open houses so that the next event can use it.
- SRS-18.6: The system shall show the number of company visits for each project.
- SRS-18.7: The system shall show the admin team a mock interview summary: students selected per department, CVs uploaded, and feedback received.
- SRS-18.8: The system shall allow the admin team to export the list of selected students and the alumni feedback to Excel and PDF.
- SRS-18.9: The system shall show the admin team the number of requirement records by status (new, matched, closed) and the number of times each student has been suggested.

19. Alumni Registration and Login

- SRS-19.1: The system shall provide a separate alumni registration link that the admin team shares only with alumni.
- SRS-19.2: The system shall allow an alumnus to register using only an email address and a password.

- SRS-19.3: The system shall not ask an alumnus for background details such as company, designation, or graduation year.
- SRS-19.4: The system shall verify that the email address is unique before creating an alumni account.
- SRS-19.5: The system shall allow an alumnus to log in immediately after registration, without admin approval.
- SRS-19.6: The system shall encrypt alumni passwords before storing them in the database.
- SRS-19.7: The system shall take an alumnus directly to the alumni dashboard after login.

20. Student Selection for Mock Interview

- SRS-20.1: The system shall allow the admin team to enter the number of alumni who will attend the mock interviews.
- SRS-20.2: The system shall allow the admin team to enter the number of students required from each department.
- SRS-20.3: The system shall allow the admin team to set the selection criteria: experience, tools, and number of projects.
- SRS-20.4: The system shall rank the students of each department using the selected criteria, based on their CV and project records.
- SRS-20.5: The system shall suggest the best students of each department up to the number entered by the admin team.
- SRS-20.6: The system shall allow the admin team to review the suggested list and add or remove students before the final selection.
- SRS-20.7: The system shall allow the admin team to enter the venue, date, and timings of the mock interview when confirming the selection.
- SRS-20.8: The system shall mark the students as selected for the mock interview when the admin team confirms the final selection.
- SRS-20.9: The system shall allow the admin team to view and export the list of selected students by department.

21. Mock Interview Notifications

- SRS-21.1: The system shall send an email and an in-app notification to every student selected for the mock interview, telling them they have been selected.
- SRS-21.2: The system shall include the venue and timings of the mock interview in the email and in the notification.
- SRS-21.3: The system shall tell the selected student in the email that attaching a CV on the portal is mandatory.
- SRS-21.4: The system shall remind selected students who have not uploaded a CV on the portal.
- SRS-21.5: The system shall notify selected students and alumni when the venue or timings change.

22. Alumni Dashboard and Student CV Access

- SRS-22.1: The system shall show the alumnus, immediately after login, a message that the admin team has selected a number of students for the mock interview and that the alumnus is to interview them.

- SRS-22.2: The system shall show the total number of selected students and the number from each department.
- SRS-22.3: The system shall list the selected students with their name, SAP ID, and program.
- SRS-22.4: The system shall allow the alumnus to open and view the CV of each selected student.
- SRS-22.5: The system shall show the alumnus the venue and timings of the mock interview.
- SRS-22.6: The system shall allow the alumnus to search the selected students by name, SAP ID, or program.
- SRS-22.7: The system shall show an alumnus only the data of the students selected for the mock interview.

23. Alumni Feedback and Review

- SRS-23.1: The system shall allow an alumnus to submit feedback for each student interviewed, with a rating and comments.
- SRS-23.2: The system shall allow an alumnus to submit overall feedback about the environment and experience of the event.
- SRS-23.3: The system shall allow an alumnus to submit feedback for the FYP students they guide.
- SRS-23.4: The system shall show a student the feedback given to them on their portal.
- SRS-23.5: The system shall allow the admin team to view the feedback of all students and the overall feedback from alumni.
- SRS-23.6: The system shall keep a record of all feedback submitted and allow the admin team to export it.

24. WhatsApp Message Intake

- SRS-24.1: The system shall allow the admin to connect his existing WhatsApp Business account to Riphah HireHive through the officially supported WhatsApp Business Platform coexistence integration, subject to account eligibility and authorization.
- SRS-24.2: The system shall allow the admin to select and approve the company contacts whose conversations are relevant to the Open House. Only selected and verified company conversations shall be eligible for storage and processing by HireHive.
- SRS-24.3: The system shall receive supported incoming and outgoing message events from selected company conversations through the configured integration, without requiring the admin to forward individual messages or manually export complete chats.
- SRS-24.4: The system shall associate messages with the relevant company using verified contact identifiers, maintain their timestamps and message direction, prevent duplicate records, and organize messages chronologically into company conversations.
- SRS-24.5: Where supported by the chosen integration and authorized during onboarding, the system shall synchronize eligible historical conversation messages. The system shall indicate when the available conversation history is incomplete or cannot be verified as complete.
- SRS-24.6: The system shall identify the company associated with a selected conversation using its verified WhatsApp contact number and existing company records. If the company identity is uncertain or its details are incomplete, the system shall request confirmation or additional information from the admin.

- SRS-24.7: The system shall analyze the available messages in a selected company conversation and extract relevant Open House recruitment requirements, including company details, required student profiles, number of positions, required skills, eligibility criteria, internship or job duration, and application deadlines where provided.
- SRS-24.8: The system shall compare the extracted information against the required company-requirement fields and identify missing, ambiguous, or unconfirmed details.
- SRS-24.9: The system shall generate an editable follow-up message for the admin to review, modify, copy, and send to the company through WhatsApp when additional information is required. The system shall not send the message automatically without an explicitly authorized messaging workflow.
- SRS-24.10: When additional messages arrive from a selected company, the system shall update the associated conversation and allow the extracted requirements to be reprocessed. Newly obtained information shall be linked to the existing requirement record rather than unnecessarily creating duplicate records.
- SRS-24.11: The system shall allow the admin to review and correct the extracted company requirements before approving them for use by subsequent system features.
- SRS-24.12: The system shall restrict message-content storage, AI processing, and requirement extraction to authorized, selected company conversations. It shall protect credentials and message data, enforce appropriate access controls, and avoid processing unrelated conversations.
- SRS-24.13: The system shall display the WhatsApp integration status, company-selection status, last synchronization time, processing status, and any synchronization or message-processing errors.
- SRS-24.14: After the admin approves the company requirements, the system shall make the verified requirement record available to Feature 25 for matching suitable students against the company's requirements.

25. Requirement-Based Student Suggestion

- SRS-25.1: The system shall match the requirements of each requirement record with the students' CVs, tools, projects, and experience.
- SRS-25.2: The system shall suggest the best matching students to the admin team for each requirement record.
- SRS-25.3: The system shall show the reason for the match next to each suggested student.
- SRS-25.4: The system shall keep a count of how many times each student has been suggested, and shall rank students who were suggested many times lower so that different students are shown for different requirements.
- SRS-25.5: The system shall still show a repeatedly suggested student when that student is the best match for the requirement.
- SRS-25.6: The system shall show on the admin dashboard each requirement record with the sender name, phone number, company, date forwarded, requirements, and the students who match.

- SRS-25.7: The system shall allow the admin team to edit the extracted details and to select students from the suggested list.
- SRS-25.8: The system shall give each requirement record a status of new, matched, or closed.
- SRS-25.9: The system shall allow the admin team to search past requirement records by sender name, company, date, or status.
- SRS-25.10: The system shall show the student suggestions only to the admin team and shall not send them to students.
- SRS-25.11: The system shall work independently of the open house and mock interview features and shall be available throughout the year.

3. Non-Functional Requirements

1. Usability

- UR-1.1: A first-time FYP student shall be able to complete team registration within 10 minutes without external help.
- UR-1.2: A first-time volunteer shall be able to complete the volunteer registration form within 5 minutes without external help.
- UR-1.3: A volunteer at the welcome desk shall be able to check in a company within 1 minute, by QR scan or by search.
- UR-1.4: A user shall be able to reach their main information (for example table number, duties, or recommended projects) within 2 interactions from the home screen.
- UR-1.5: The system shall prevent a form from being submitted when required fields are empty, show an error message that explains how to fix it (for example the correct Riphah email format), and show a confirmation after every successful submission.
- UR-1.6: The system shall use simple words, the same terms, and the same navigation on all screens, allow text size to be increased up to 200%, and not use color alone to show important information.
- UR-1.7: At least 80% of representative test users shall rate the system as satisfactory or higher in a post-test survey.
- UR-1.8: A first-time alumnus shall be able to register within 2 minutes without external help.
- UR-1.9: An alumnus shall be able to submit the feedback form for one student within 3 minutes.
- UR-1.10: The admin team shall be able to open the suggested students for a forwarded message within 2 interactions from the dashboard.

2. Performance

- PR-1.1: Each page shall load within 3 seconds under normal use.
- PR-1.2: Search results for CVs, projects, and companies shall appear within 3 seconds.
- PR-1.3: A check-in record shall be saved within 2 seconds.
- PR-1.4: The system shall support at least 300 users at the same time on the event day without errors.
- PR-1.5: The system shall generate table tags for all projects within 1 minute.
- PR-1.6: The system shall send a notification or email to a whole group within 5 minutes.
- PR-1.7: The system shall show the ranked list of suggested students for the mock interview within 5 seconds.
- PR-1.8: The alumni dashboard, with the student list, shall load within 3 seconds.
- PR-1.9: The system shall scan a forwarded message and save its requirement record within 30 seconds.
- PR-1.10: The system shall show the suggested students for a requirement record within 5 seconds.

3. Security

- SEC-1.1: The system shall authenticate every user before giving access to any data.
- SEC-1.2: The system shall store passwords in encrypted form only.
- SEC-1.3: The system shall lock an account for a short time after 5 failed login attempts.

- SEC-1.4: The system shall not allow a user to open another role's screens or data, even by changing the web address.
- SEC-1.5: The system shall show a student's CV and phone number only to authorized users, and to companies only when the student has allowed it.
- SEC-1.6: The system shall transfer all data over an encrypted connection (HTTPS).
- SEC-1.7: The system shall log out a user automatically after 30 minutes of inactivity.
- SEC-1.8: The system shall accept only allowed file types (for example PDF and Word) and a maximum file size when users upload CVs and documents.
- SEC-1.9: The system shall use student and company data only for open house, mock interview, and requirement-matching purposes.
- SEC-1.10: The system shall show an alumnus only the details and CVs of the students selected for the mock interview.
- SEC-1.11: The system shall show a student's feedback only to that student and to the admin team.
- SEC-1.12: The system shall show forwarded WhatsApp messages, sender phone numbers, and requirement records only to the admin team.
- SEC-1.13: The system shall store the WhatsApp connection credentials in encrypted form and shall use only the official WhatsApp Business connection.

4. Reliability and Availability

- REL-1.1: The system shall be available at least 99% of the time during the registration period and on the event day.
- REL-1.2: The system shall take an automatic backup of all data every day.
- REL-1.3: The system shall be able to restore data from the latest backup within 2 hours.
- REL-1.4: The system shall not lose already saved data if the internet or server fails.
- REL-1.5: The system shall show a clear error message and keep working when a service such as email is unavailable.
- REL-1.6: The system shall not create duplicate records when a user submits the same form twice.
- REL-1.7: The system shall let the admin team continue by pasting a message manually and shall show a clear error message when the WhatsApp connection is not available.
- REL-1.8: The system shall not create duplicate requirement records or send the same template message twice when a message is forwarded or approved twice.

5. Compatibility and Portability

- COM-1.1: The system shall work on the latest versions of Chrome, Edge, Firefox, and Safari.
- COM-1.2: The system shall adjust its screens to phone, tablet, and laptop sizes.
- COM-1.3: The exported Excel and PDF files shall open correctly in standard Excel and PDF readers.
- COM-1.4: The printable table tags and PDFs shall print correctly on A4 paper.
- COM-1.5: The generated QR codes shall be readable by a standard phone camera.

6. Scalability

- SCA-1.1: The system shall allow the number of FYP projects, volunteers, and companies to grow in later open houses without redesign.

- SCA-1.2: The system shall store the data of multiple open houses separately.
- SCA-1.3: The table numbering shall keep working correctly when more programs or projects are added.
- SCA-1.4: The system shall store the mock interview selections and feedback of each open house separately.

7. Maintainability

- MNT-1.1: The admin team shall be able to change deadlines, email templates, and the evaluation rubric without changing the code.
- MNT-1.2: The system shall be built in separate modules so that one module can be updated without affecting the others.
- MNT-1.3: The system shall record errors in a log for troubleshooting.
- MNT-1.4: The project shall provide a user manual for the admin team and volunteers.
- MNT-1.5: The admin team shall be able to change the selection criteria and the feedback form fields without changing the code.
- MNT-1.6: The admin team shall be able to change the matching rules and the text of the WhatsApp message template without changing the code.